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BUILDING A FILE SHARING APPLICATION

Network Programming - IT4062E - 1st Progress Report

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ONE LOVE. ONE FUTURE.

1. Actors & Problem Definition

System Actors

- **Regular Member**:
  - Registers & signs in to the system.
  - Requests to join groups.
  - Uploads/downloads files, creates subfolders in group space.
- **Group Admin** (Group Creator):
  - Creates and manages sharing groups.
  - Exclusive rights: rename & delete files and subfolders.

Problem Definition

- **Group-Centric Collaboration**: Provide dedicated shared folders for teams to organize and collaborate online.
- **Hierarchical Content Organization**: Allow members to structure project data with nested subfolders and file uploads.
- **Role-Based Access Control**: Enforce administrative authority so that only group admins can rename or delete files and folders, preventing unauthorized data loss.
- **Network File Transfer**: Support secure client-server file uploads and downloads across the network.

2. Core Use Cases

General Use Cases

- **UC1 — Register & Log In**
  Create an account and authenticate for usage.
- **UC2 — Create or Join Group**
  Create a new group or request to join existing ones.
- **UC3 — Upload & Download Files**
  Upload files to group storage and download shared files.
- **UC4 — Create Subfolders**
  Create subdirectories within the group folder.

Admin-Only Use Cases

- **UC5 — Delete Files & Subfolders**
  Exclusive authority to remove files and nested folders within the group space.
- **UC6 — Rename Files & Folders**
  Exclusive authority to rename files and directories.
- **UC7 — Manage Group Membership**
  Review, accept, or reject user requests to join the group.

3. Technology Stack & Development Phases

Technology Stack

- **Core Language: Java (JDK 17+)**
  Object-oriented design, cross-platform portability, robust standard I/O.
- **Transport Protocol: TCP Sockets**
  Connection-oriented `java.net` sockets ensuring reliable, lossless byte streams.
- **Concurrency: Multithreaded Engine**
  Thread pools (`ExecutorService`) to handle concurrent client sessions.

General Development Phases

Phase 1: Protocol Framing & Storage Schema
Custom packet format & per-group directory layout

Phase 2: TCP Socket Engine & Authentication
Multithreaded socket server, connection handling, register & login

Phase 3: Group Management & RBAC
Group creation, member join requests, and admin privilege controls

Phase 4: File Transfer & Directory Operations
Streaming upload/download, subfolder creation, rename & delete

Phase 5: Integration & Concurrency QA
Multi-client stress testing, network error recovery & verification



THANK YOU !